Year 2 Pure Mathematics

Chapter 8

Parametric equations

Practice Booklet

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8.1 Parametric equations

Practice Book, printed page 60.

1. For each of the following parametric equations, find a Cartesian equation, giving your answer in the form $y = \mathrm{f}(x)$. In each case find the domain and range of $\mathrm{f}(x)$.

a. $x = t - 3$, $y = t^2 - 1$, $-3 \leq t \leq 3$

b. $x = \dfrac{2}{t}$, $y = t^2 + 1$, $t > 0$

c. $x = 2t - 1$, $y = \dfrac{3}{t^2}$, $t \geq 1$

Hint

A Cartesian equation in two dimensions involves the variables x and y only. Rearrange one parametric equation into the form $t = \ldots$, and then substitute into the other equation to eliminate t.

2. For each of the following parametric curves, find a Cartesian equation, giving your answer in the form $y = \mathrm{f}(x)$. In each case find the domain and range of $\mathrm{f}(x)$.

a. $x = \ln(t + 2)$, $y = \dfrac{1}{t + 3}$, $t > -1$

b. $x = \ln(4 + t)$, $y = t^2 + 10t + 25$, $t > -3$

c. $x = e^{2t}$, $y = e^{6t} + 2e^{4t}$, $t \in \mathbb{R}$

Hint

To find the domain of $\mathrm{f}(x)$, consider the range of values of x for the given domain of t.
To find the range of $\mathrm{f}(x)$, consider the range of values of y for the given domain of t.

3. For each of these parametric curves:

i. find a Cartesian equation for the curve in the form $y = \mathrm{f}(x)$

ii. find the domain and range of $\mathrm{f}(x)$

iii. sketch the curve within the given domain of t.

a. $x = t^{\frac{1}{2}} + 1$, $y = 2t^{\frac{1}{2}} + 7$, $0 < t \leq 16$

b. $x = 2t + 1$, $y = 4t^2 + 14t + 12$, $-3 \leq t \leq 0$

c. $x = \ln(2 - t)$, $y = 5 - t$, $t < 1$

4. E The parametric equations $x = 3t$, $y = (1 - t)(9 - t)$, $0 \leq t \leq 8$ define a curve A.

a. Find the Cartesian equation of the curve in the form $y = \mathrm{f}(x)$, and state the domain and range of $\mathrm{f}(x)$. (3 marks)

b. Sketch the curve A, clearly showing the coordinates of any points of intersection with the axes, turning points or endpoints. (3 marks)

5. E/P A curve C is defined by the parametric equations $x = \ln(t+5)$, $y = 2t + 12$, $t \geq -4$.

a. Find the Cartesian equation of the curve in the form $y = \mathrm{f}(x)$. (2 marks)

b. Write down the minimum value of x and the minimum value of y. (2 marks)

6. E/P A curve M has parametric equations $x = e^{3t}$, $y = e^{9t} - 2e^{6t} - 5e^{3t} + 6$, $t \in \mathbb{R}$.

a. Show that the Cartesian equation of M can be written in the form $y = (x+a)(x-b)(x-c)$, where a, b, and c are integers to be determined. (4 marks)

b. Hence write down the coordinates of the points where the curve crosses each axis. (3 marks)

7. E/P The curves C_1 and C_2 are defined parametrically as follows:

$$C_1 : x = 5t^2,\ y = 3t^2 + 2,\ 0 \leq t \leq 2$$

$$C_2 : x = 3\sqrt{t} - 3,\ y = 9 - 5\sqrt{t},\ 0 \leq t \leq 4$$

a. Show that the curves are line segments which are perpendicular to each other. (3 marks)

b. Find the exact length of each line segment. (2 marks)

8. E/P The curve C has parametric equations $x = 4t^2$, $y = 8t^3$, $-1 \leq t \leq 1$.
Find a Cartesian equation of C in the form $y^2 = \mathrm{f}(x)$. (4 marks)

8.2 Using trigonometric identities

Practice Book, printed page 61.

1. Find the Cartesian equations of the curves given by the following parametric equations:

a. $x = \sin t - 1$, $y = \cos t + 3$, $0 < t < 2\pi$

b. $x = 2\sin t$, $y = 3\cos t$, $0 < t < \pi$

c. $x = 2 + 2\cos t$, $y = 5 + 2\sin t$, $0 < t < 2\pi$

Hint

Use $\sin^2 t + \cos^2 t \equiv 1$. You do not need to give the equations in the form $y = \mathrm{f}(x)$.
← Year 1, Section 10.3

2. Find the Cartesian equations of the curves given by the following parametric equations:

a. $x = \sin t$, $y = \cos 2t$, $0 < t < 2\pi$

b. $x = 4\sec t$, $y = 2\tan t$, $0 < t < 2\pi$

c. $x = \sin t$, $y = \sin^2 2t$, $0 < t < 2\pi$

Hint

$\cos 2t \equiv 1 - 2\sin^2 t$ ← Section 7.3
$1 + \tan^2 t \equiv \sec^2 t$ ← Section 6.4
$\sin 2t \equiv 2\sin t \cos t$ ← Section 7.3

3. For each of these parametric curves:

i. find a Cartesian equation in the form $y = \mathrm{f}(x)$

ii. sketch the curve within the given domain of t.

a. $x = \sin t + 3$, $y = \cos t - 4$, $0 < t < 2\pi$

b. $x = \dfrac{2}{\tan t}$, $y = 4\,\mathrm{cosec}^2 t - 8$, $0 < t < \pi$

c. $x = 10\sin t$, $y = \dfrac{1}{5}\,\mathrm{cosec}^2 t$, $0 < t < \dfrac{\pi}{2}$

Hint

$(x-a)^2 + (y-b)^2 = r^2$ is the equation of a circle with centre (a, b) and radius r.
← Year 1, Section 6.2

4. P A curve C has parametric equations

$$x = 6\cos t + 5, \quad y = 6\sin t - 2, \quad 0 \leq t \leq k$$

When $k = 2\pi$, C is a circle.

a. Write a Cartesian equation for the circle, and hence state its radius and the coordinates of its centre.

b. Write down a value of k such that C is a semicircle.

5. E/P A curve C has parametric equations

$$x = 5\sec^2 2t, \quad y = 2\cot^2 2t, \quad 0 < t \leq \frac{\pi}{4}$$

a. Find a Cartesian equation of the curve, expressing y in terms of x. (5 marks)

b. State the range of possible values of x for the given domain of t. (1 mark)

6. E/P Show that the curve with parametric equations $x = 2\sin t$, $y = \cos\left(t + \frac{\pi}{3}\right)$, $-\frac{\pi}{2} < t < \frac{\pi}{2}$, can be written in the form $y = \frac{1}{4}(\sqrt{4 - x^2} - \sqrt{3}x)$, $-k < x < k$, where k is a real constant to be found. (6 marks)

7. E/P A curve has parametric equations

$$x = \cot^2 t + 3, \quad y = 3\cos t, \quad 0 < t < \frac{\pi}{2}$$

a. Find the Cartesian equation of the curve in the form $y = \mathrm{f}(x)$. (4 marks)

b. Find the domain and range of $y = \mathrm{f}(x)$ in the given domain of t. (2 marks)

8.3 Curve sketching

Practice Book, printed page 62.

1. A curve is given by the parametric equations

$$x = 2t + 2, \quad y = t^2, \quad -2 \leq t \leq 3$$

Copy and complete the table and draw a graph of the curve.

t	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3
$x = 2t + 2$	-2	-1			2						
$y = t^2$	4	2.25		0.25							

Hint

For each value of t, substitute into the parametric equations to find the corresponding values of x and y.

2. A curve is given by the parametric equations

$$x = 3\sin t, \quad y = 5\cos t, \quad 0 \leq t \leq 2\pi$$

Copy and complete the table and draw a graph of the curve for $0 \leq t \leq 2\pi$.

t	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$x = 3\sin t$	0	2.12							
$y = 5\cos t$	5	3.54							

Hint

Make sure you plot x against y. Do not plot any values of t.

3. By using a table of values, or otherwise, sketch the curves given by these parametric equations:

a. $x = 2t^2$, $y = 4t$, $-2 \leq t \leq 2$

b. $x = 3\operatorname{cosec} t$, $y = 3\cot t$, $0 \leq t \leq 2\pi$

c. $x = t + 5$, $y = \dfrac{1}{t}$, $0 \leq t \leq 8$

4. P The curves C_1, C_2 and C_3 are defined parametrically as follows:

$$C_1 : x = 2t,\ y = 2.5 - t,\ 0.5 \leq t \leq 2.5$$

$$C_2 : x = \frac{1}{t},\ y = \frac{2}{t},\ \frac{1}{3} \leq t \leq 1$$

$$C_3 : x = 2t + 1,\ y = 12 - 6t,\ 1 \leq t \leq 2$$

a. Find Cartesian equations of C_1, C_2 and C_3 in the form $y = \mathrm{f}(x)$.

b. On the same set of coordinate axes, sketch C_1, C_2 and C_3 on their given domains.

c. Find the area of the shape enclosed by C_1, C_2 and C_3.

5. E/P The curve C is defined by the parametric equations

$$x = t + 2, \quad y = -5t - t^2, \quad -5 \leq t \leq 1$$

a. Find a Cartesian equation of C in the form $y = (a - x)(x + b)$, where a and b are integers to be found. (4 marks)

b. Sketch the curve C on the given domain of t. (3 marks)

6. E/P The curve C is defined by the parametric equations

$$x = 10\cos t + 3, \quad y = 10\sin t - 2, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

a. Find a Cartesian equation of C in the form $(x + a)^2 + (y + b)^2 = c$, stating the values of a, b and c. (4 marks)

b. Sketch the curve C on the appropriate domain. (3 marks)

c. Find the exact length of the curve C. (3 marks)

8.4 Points of intersection

Practice Book, printed page 63.

1. Find the coordinates of the point(s) where the following curves meet the x-axis and the y-axis.

a. $x = t - 3$, $y = t + 4$, $\in \mathbb{R}$

b. $x = 3t$, $y = 3(t + 2)(t - 1)$, $\in \mathbb{R}$

c. $x = 2t^2 - 2$, $y = t^2 - 4t$, $\in \mathbb{R}$

Hint

Substitute $y = 0$ or $x = 0$ into the parametric equations to find the value(s) of t when the curve crosses the x-axis and the y-axis respectively. Then find the corresponding coordinates.

2. Find the coordinates of the point(s) where the following curves meet the x-axis and the y-axis.

a. $x = \sin 2t + 1$, $y = 2\tan^2 t$, $-\dfrac{\pi}{4} \leq t \leq \dfrac{\pi}{4}$

b. $x = 2t$, $y = \cos t$, $-\pi \leq t \leq \pi$

c. $x = e^t - 3$, $y = e^{2t} - 2$, $t > 0$

Hint

The curve in part b intersects the x-axis twice within the given domain of t.

3. The line L is defined parametrically by

$$x = t + 1, \quad y = t + 3, \quad t \in \mathbb{R}$$

The circle C has equation $x^2 + y^2 + 6x - 4 = 0$.

a. Find the values of t at the points of intersection of L and C.

b. Hence find the coordinates of these points.

Hint

Substitute the expressions for x and y in the parametric equations for L into the equation for C, then solve the resulting quadratic equation to find two possible values for t.

4.

a. A curve has parametric equations $x = p(1 + 2t)$, $y = p(1 - 2t)^2$, $t \in \mathbb{R}$ where p is a constant. The curve passes through the point $(6, 0)$. Find the value of p.

b. A curve has parametric equations $x = qt^2 - 2q$, $y = q(t-1)(t-3)^2$, $t > 1$, where q is a constant. The curve passes through the point $\left(\frac{7}{2}, 0\right)$. Find the value of q.

5. E/P A curve C has parametric equations $x = e^{2t}$, $y = 4e^t - 4$, $t \in \mathbb{R}$.

a. Find the values of t where the line $y = x - 1$ crosses the curve C. (3 marks)

b. Hence determine the coordinates of the points where the line $y = x - 1$ intersects the curve C. (2 marks)

6. E/P A curve C is defined by the parametric equations

$$x = \cos t, \quad y = 2\sin 2t + 1, \quad \frac{\pi}{4} < t < \frac{3\pi}{2}$$

a. Determine the coordinates of the points where the line $y = 1$ crosses the curve C. (5 marks)

b. Show that the curve cuts the x-axis where $t = \frac{7\pi}{12}$ and $t = \frac{11\pi}{12}$ (3 marks)

c. Hence determine the exact coordinates of the points where the curve cuts the x-axis. (2 marks)

7. E/P A curve C has parametric equations $x = 4\sin t$, $y = 8\cos t$, $0 < t < 2\pi$.

The points P and Q lie on C where $t = \frac{\pi}{6}$ and $t = \frac{3\pi}{2}$ respectively.

Find an equation of the straight line through P and Q, giving your answer in the form $ay + bx + c = 0$. (5 marks)

8. E/P A curve has parametric equations $x = \ln(t + 2)$, $y = \ln(t - 1)$, $t > 1$.

a. Show that the line $y = 2x - \ln 18$ crosses the curve at the points where $t = a \pm b\sqrt{3}$, where a and b are integers to be found. (4 marks)

b. Hence determine the exact coordinates of the points where the line $y = 2x - \ln 18$ intersects the curve. (3 marks)

8.5 Modelling with parametric equations

Practice Book, printed page 65.

1. P A ski lift lifts a skier from ground level to an elevation of 450 m.
The position of the skier at time t seconds is modelled by the parametric equations

$$x = 3.6t \text{ m}, \quad y = 2.7t \text{ m}$$

where x is the distance travelled horizontally and y is the distance travelled vertically.

a. Find the time taken to travel to the top of the ski lift.

b. Show that the motion of the skier is a straight line.

c. Hence find the total distance travelled by the skier in travelling from ground level to the top of the ski lift.

d. Determine the speed of the ski lift.

e. Criticise the model in relation to:

i. the path of the skier

ii. the speed of the ski lift.

2. P Adam's position as he travels down a water slide at time t can be modelled by the parametric equations $x = 8t$, $y = 16 - 7.6t - \cos 8t$, $0 < t < k$, where x is the horizontal distance travelled (in metres) and y is Adam's vertical height above ground level (in metres).

a. Find Adam's initial height above ground level.

b. Find Adam's vertical height above ground level after $\frac{17}{8}$ seconds.

c. State the horizontal distance that Adam has travelled after $\frac{17}{8}$ seconds, giving your answer to the nearest metre.

3. P A ball is kicked from the ground with an initial speed of $18\,\text{m}\,\text{s}^{-1}$ at an angle of $45°$.
Its position after t seconds can be described using the parametric equations

$$x = 9\sqrt{2}t\,\text{m}, \quad y = (-kt^2 + 9\sqrt{2}t)\,\text{m}, \text{ where } k \text{ is a constant}$$

a. Given that ball travels a horizontal distance of 33 m before hitting the ground, find the time of flight of the ball.

b. Hence find the value of k.

c. Show that the path of the ball is modelled by a quadratic curve.

4. P The path of a kangaroo as it jumps can be modelled by the parametric equations

$$x = 5.6t\,\text{m}, \quad y = -4.9t^2 + 2.1t\,\text{m}$$

where x is the horizontal distance from the point the kangaroo jumps off the ground, y is the height above the ground and t is the time in seconds after the kangaroo has started its jump.

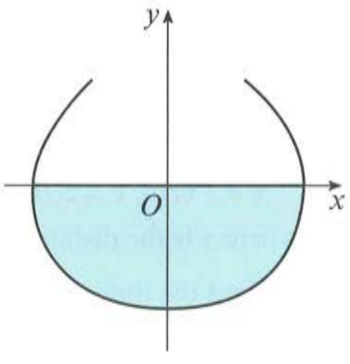
a. Find the time the kangaroo takes to complete a single jump.

b. Find the horizontal distance the kangaroo travels during a single jump.

c. The kangaroo jumps over a log that is 20 cm high. Find the range of the distances from the log that the kangaroo must jump from in order to clear it.

5. E/P The cross-section of a vase design is given by the parametric equations

$$x = (18t - 32\sin t)\,\text{cm}, \quad y = (8 - 16\cos t)\,\text{cm}, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$



a. Find the width of the opening of the vase. (3 marks)

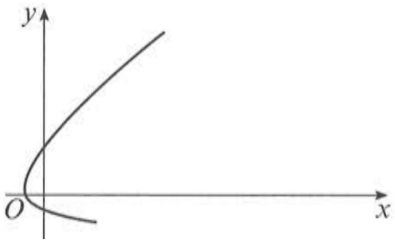
b. The vase is filled with water up to the level of the x-axis. Find the radius of the vase at the surface of the water. (3 marks)

c. Given that the cross-section of the vase crosses the y-axis at its deepest point, find the depth of the water in the vase. (3 marks)

6. E/P Particle A is moving in the xy-plane such that its position relative to a fixed origin O at time t seconds is given by the parametric equations

$$x = \frac{t^2 - 6t + 5}{t + 1}, \quad y = t - 2, \quad t \geq 0$$

The diagram shows the path of the particle.



Particle B is moving in the xy-plane such that its position relative to the same fixed origin O at time t seconds is given by the parametric equations

$$x = t + 5, \quad y = 5t - 2, \quad t \geq 0$$

a. Show that the path of particle B is a straight line. (2 marks)

Both particles are at $(5, -2)$ when $t = 0$.

b. Find the exact coordinates of the other point where the paths of the particles intersect. (3 marks)

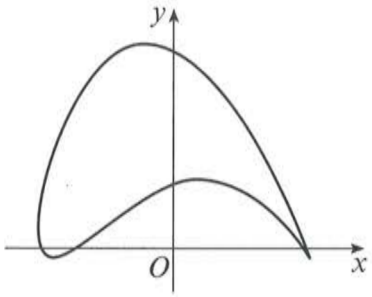
Adam states that the particles will collide at this point.

c. State whether you agree with Adam's statement, giving a reason for your answer. (2 marks)

7. E/P A walker travels around one loop of a path in a park. The position of the walker relative to the entrance of the park, O, at time t minutes, is modelled using the parametric equations

$$x = 400\cos\left(\frac{1}{5}\left(t - \frac{\pi}{3}\right)\right), \quad y = 400\sin^2\frac{t}{5} - 200\sin\frac{t}{5}, \quad t \geq 0$$

where x and y are measured in metres, and the positive y-axis is the north direction.

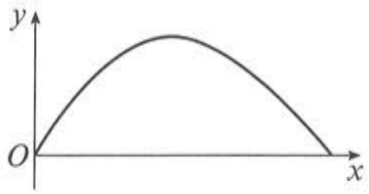


a. Find the distance of the walker from the park entrance at the beginning of his motion. (3 marks)

b. Find the distances of the walker from the park entrance at the times when he is due north of the park entrance. (4 marks)

c. Determine how long it takes the walker to complete one loop of the path to the nearest minute. (3 marks)

8. E/P The diagram shows the profile of a hill modelled by the parametric equations $x = t^2 + 16t$, $y = 20\sin t$, $0 < t < k$, where x represents the distance in metres measured horizontally from a fixed point, and y represents the height above sea level in metres.



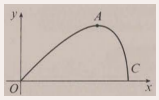
a. Find the height of the hill. (3 marks)

b. Find the value of k and hence find the average gradient of the descent away from the origin. (4 marks)

Problem solving: Set A

Bronze

A curve C has parametric equations
$x = 10\sin t,\ y = 5\sin 2t,\ 0 \le t \le \frac{\pi}{2}$



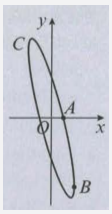
a. **i.** Show that $y = x\cos t$.
ii. Hence show that a Cartesian equation of C can be written as $100(y^2 - x^2) + x^4 = 0$. (4 marks)

The point A lies on C at the point where $t = \frac{\pi}{4}$
The line with equation $y = x + k$, where k is a constant, intersects the curve at point A.

b. Find:
i. the coordinates of point A
ii. the exact value of k. (4 marks)

Silver

A curve C has parametric equations
$x = \frac{2}{\sqrt{3}}\sin t,\ y = 4\cos\left(t + \frac{\pi}{3}\right),\ 0 \le t \le 2\pi$



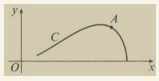
a. Show that all points on C satisfy $(3x + y)^2 + 3x^2 - 4 = 0$. (5 marks)

A straight line l passes through the points A and B on curve C where $t = \frac{\pi}{6}$ and $t = \frac{\pi}{2}$ respectively.

b.
i. Show that the gradient of line l is -6.
ii. Hence find the equation for line l in the form $ax + by + c = 0$. (4 marks)

Gold

A curve C has parametric equations
$x = 2 + 2\cos t,\ y = 2\sin t + \sin 2t,\ 0 \le t \le \frac{3\pi}{4}$



a. Show that all points on C satisfy $y^2 = x^3\left(1 - \frac{1}{4}x\right)$, $a \le x \le b$, stating the exact values of a and b. (6 marks)

The line with equation $x + y = p$, where p is a constant, is a tangent to C at point A, where $t = \frac{\pi}{4}$

b. Find the value of p. (2 marks)

c. Given that the line $x + y = q$ intersects curve C at least once, find the exact range of possible values of q, writing your answer in set notation. (2 marks)

Problem solving: Set B

Bronze

The motion of a football as it is kicked is modelled using the parametric equations
$x = 7.6t,\ y = -4.9t^2 + 16.3t$
where x m is the horizontal distance travelled and y m is the height of the ball above the ground after t seconds.

a. Find the two values of x for which the football is exactly 10 m above the ground. (4 marks)

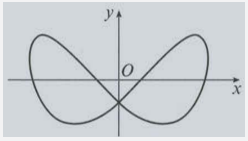
b. Show that the path of the ball is a parabola and work out the maximum height of the ball above the ground. (5 marks)

Silver

The motion of a dancer relative to a fixed origin O at time t minutes is modelled using the parametric equations

$x = 16\sin 4t,\ y = 32\cos\left(8t - \frac{8\pi}{3}\right),\ t \geq 0$

where x and y are measured in metres.



a. Find the distance of the dancer from the origin at time $t = 1$. (2 marks)

b. Find the coordinates of the point where the dancer intersects his own path. (4 marks)

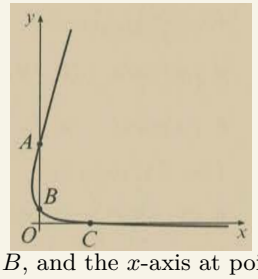
c. Determine how long it takes the dancer to complete one complete figure-of-eight motion. (4 marks)

Gold

A particle is moving in the xy-plane such that its position after time t seconds relative to a fixed origin O is given by the parametric equations

$x = \frac{t^2 - 7t + 10}{t},\ y = t\ln t,\ t \geq 0$

The diagram shows the path of the particle.



The particle crosses the y-axis at points A and B, and the x-axis at point C as shown on the diagram.

a. Find the area of triangle ABC, giving your answer in the form $\ln k$ where k is a rational number to be found. (5 marks)

b. Given that the line with equation $x = k$ intersects the curve exactly once, determine the value of k. (3 marks)

Now try this ⟶ Exam question bank Q10, Q31, Q34, Q45, Q48, Q54, Q72, Q86

Selected exam question bank questions

Question 10

Show that the line with equation $y = 3x + 2$ does not intersect the curve with parametric equations $x = t + 1$, $y = (2t + 2)(2 - t)$, $t \in \mathbb{R}$. (4 marks)

Question 31

The curve C has parametric equations $x = 5\cos t - 4$, $y = 5\sin t + 1$, $-\frac{\pi}{2} \leq t \leq \frac{\pi}{4}$

a. Find a Cartesian equation of C in the form $(x + a)^2 + (y + b)^2 = c$, stating the values of a, b and c. (4 marks)

b. Sketch the curve C on the appropriate domain. (3 marks)

c. Determine the exact length of curve C. (2 marks)

Question 34

The motion of a netball as it leaves a player's hand and passes through the net is modelled using the parametric equations $x = 3.8t$, $y = -4.9t^2 + 4.6t + 1.9$, $t \geq 0$, where x m is the horizontal distance travelled and y m is the height of the netball above the ground after t seconds.

a. Write down the height of the ball when it leaves the player's hand. (1 mark)

Given that the height of the net is 2.8 m and that the netball passes through the net as it is travelling downwards,

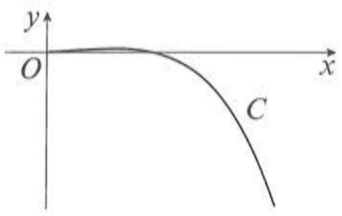
b. find the time at which the netball passes through the net (3 marks)

c. find the horizontal distance between the player and the net. (1 mark)

Question 45

The diagram shows the curve C which is defined by the parametric equations

$$x = t^{\frac{1}{2}}, \quad y = 2t^{\frac{1}{2}} - 4t^2, \quad 0 \leq t \leq 3$$

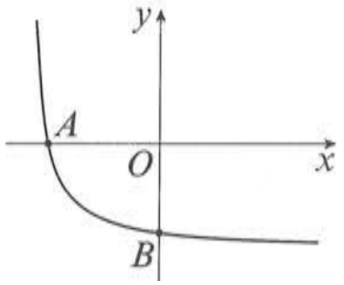


a. Find a Cartesian equation of the curve in the form $y = \mathrm{f}(x)$ and state the domain of $\mathrm{f}(x)$. (3 marks)

b. Find the value of t when $\frac{\mathrm{d}y}{\mathrm{d}t} = 0$, and hence determine the exact range of $\mathrm{f}(x)$. (4 marks)

Question 48

The curve C has parametric equations $x = t - 2$, $y = \dfrac{1 - 3t}{t}$, $t > 0$.

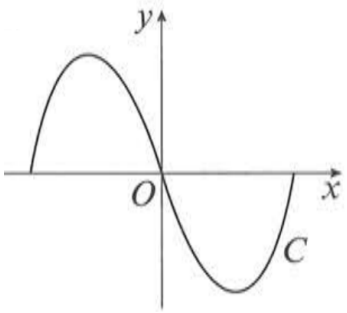


a. The curve crosses the x-axis at point A and the y-axis at point B. Find the coordinates of A and B. (4 marks)

b. A straight line l intersects the curve C at points A and B.
Find the equation of the line l in the form $ax + by + c = 0$. (3 marks)

Question 54

A curve C has parametric equations $x = \dfrac{1}{6}\cos t$, $y = \cos 3t$, $\dfrac{\pi}{6} \leq t \leq \dfrac{5\pi}{6}$



a. Show that the Cartesian equation of the curve can be written in the form $y = ax(bx^2 - c)$, stating the values of a, b and c. (5 marks)

b. Determine the range of possible values of x in the given domain of t. (1 mark)

c. Find the coordinates of the turning points of the curve, and hence determine the range of possible values of y in the given domain of t. (3 marks)

Question 72

A curve C is defined by the parametric equations $x = e^{2t}$, $y = 6e^t - 2$, $t \in \mathbb{R}$.

a. The straight line $y = x + 3$ passes through the points A and B on curve C.
Find the coordinates of A and B. (6 marks)

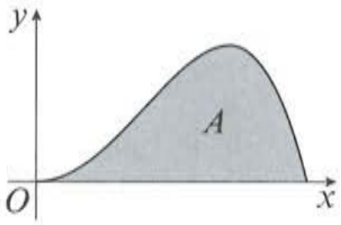
b. Another straight line, l, passes through the points P and Q on curve C where $t = \ln 2$ and $t = \ln 3$ respectively. Find an equation for line l in the form $ax + by + c = 0$. (6 marks)

Question 86

The position of a carriage on a rollercoaster at time t seconds is modelled using the parametric equations

$$x = 40\sqrt{t}, \quad y = 80t - 20t^2, \quad 0 \leq t \leq T$$

where x represents the distance travelled horizontally in metres and y represents the height above the ground in metres.



Given that the rollercoaster starts and finishes at ground level,

a. find the value of T (2 marks)

b. find the maximum height of the rollercoaster and the time at which the carriage reaches this height. (3 marks)

c.

i. Find the angle of descent at the point where the rollercoaster reaches the ground at the end of its path.

ii. Hence comment on the validity of the model.

(4 marks)

d. Find the total area bounded by the path of the rollercoaster and the ground, labelled A on the diagram. (3 marks)

Answers

8.1 Parametric equations

1.

a. $y = (x+3)^2 - 1,\ -6 \le x \le 0,\ -1 \le y \le 8$

b. $y = \dfrac{4}{x^2} + 1,\ x > 0,\ y > 1$

c. $y = \dfrac{12}{(x+1)^2},\ x \ge 1,\ 0 < y \le 3$

2.

a. $y = \dfrac{1}{e^x + 1},\ x > 0,\ 0 < y < \dfrac{1}{2}$

b. $y = e^{2x} + 2e^x + 1,\ x > 0,\ y > 4$

c. $y = x^3 + 2x^2,\ x > 0,\ y > 0$

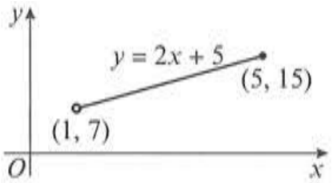
3.

a.

i. $y = 2x + 5$

ii. $1 < x \le 5,\ 7 < y \le 15$

iii.

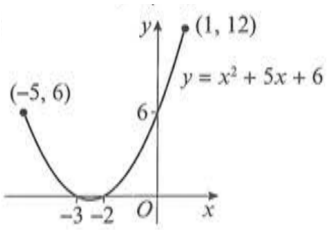


b.

i. $y = x^2 + 5x + 6$

ii. $-5 \le x \le 1,\ -\dfrac{1}{4} \le y \le 12$

iii.

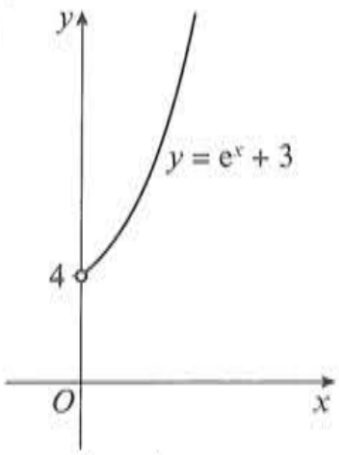


c.

i. $y = e^x + 3$

ii. $x > 0,\ y > 4$

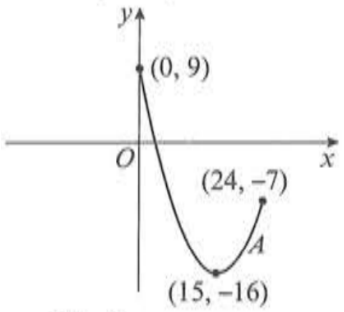
iii.



4.

a. $y = 9 - \frac{10}{3}x + \frac{1}{9}x^2$, $0 \leq x \leq 24$, $-16 \leq y \leq 9$

b.



5.

a. $y = 2e^x + 2$

b. $x = 0$, $y = 4$

6.

a. $y = x^3 - 2x^2 - 5x + 6 \equiv (x + 2)(x - 1)(x - 3)$

b. x-axis: $(-2, 0)$, $(1, 0)$, $(3, 0)$, y-axis: $(0, 6)$

7.

a. C_1: $x = 5t^2$, $y = 3t^2 + 2$,
$\Rightarrow y = 3\left(\frac{x}{5}\right) + 2 \Rightarrow y = \frac{3}{5}x + 2$
C_1 is a straight line, gradient $\frac{3}{5}$
C_2: $x = 3\sqrt{t} - 3$, $y = 9 - 5\sqrt{t}$,
$\Rightarrow y = 9 - 5\left(\frac{x+3}{3}\right) \Rightarrow y = 4 - \frac{5}{3}x$
C_2 is a straight line, gradient $-\frac{5}{3}$
$\frac{3}{5} \times -\frac{5}{3} = -1$, therefore the line segments are perpendicular.

b. C_1: $4\sqrt{34}$, C_2: $2\sqrt{34}$

8. $y^2 = x^3$

8.2 Using trigonometric identities

1.

a. $(x + 1)^2 + (y - 3)^2 = 1$

b. $\frac{x^2}{4} + \frac{y^2}{9} = 1$

c. $(x - 2)^2 + (y - 5)^2 = 4$

2.

a. $y = 1 - 2x^2$

b. $\frac{x^2}{16} - \frac{y^2}{4} = 1$

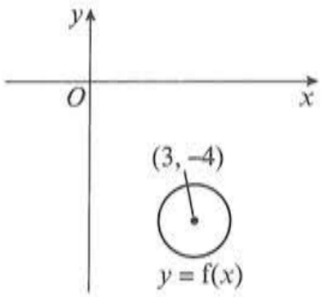
c. $y = 4x^2 - 4x^4$

3.

a.

i. $(x - 3)^2 + (y + 4)^2 = 1$

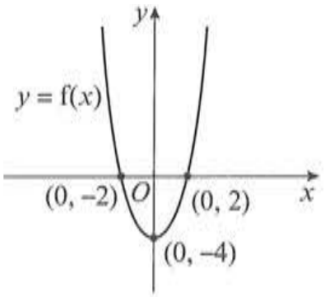
ii.



b.

i. $y = x^2 - 4$

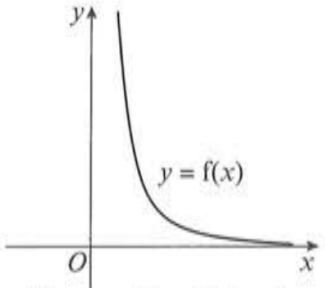
ii.



c.

i. $y = \dfrac{20}{x^2}$

ii.



4.

a. $(x-5)^2 + (y+2)^2 = 36$. Radius $= 6$, centre $(5, -2)$

b. π

5.

a. $y = \dfrac{10}{x-5}$

b. $x > 0$

6. $x = 2\sin t \Rightarrow \dfrac{x}{2} = \sin t \Rightarrow \dfrac{x^2}{4} = \sin^2 t$

$y = \cos\left(t + \dfrac{\pi}{3}\right) \Rightarrow y = \dfrac{1}{2}\cos t - \dfrac{\sqrt{3}}{2}\sin t$

$\Rightarrow y = \dfrac{1}{2}\sqrt{1 - \sin^2 t} - \dfrac{\sqrt{3}}{2}\left(\dfrac{x}{2}\right)$

$\Rightarrow y = \dfrac{1}{2}\sqrt{1 - \dfrac{x^2}{4}} - \dfrac{\sqrt{3}x}{4}$

$\Rightarrow y = \dfrac{1}{2} \times \dfrac{\sqrt{4 - x^2}}{2} - \dfrac{\sqrt{3}x}{4}$

$\Rightarrow y = \dfrac{1}{4}(\sqrt{4 - x^2} - \sqrt{3}x),\ -2 < x < 2$

7.

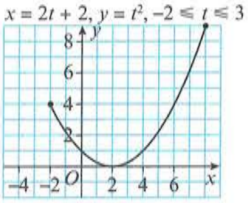
a. $y = 3\sqrt{\dfrac{x-3}{x-2}}$

b. $x > 3,\ 0 < y < 3$

8.3 Curve sketching

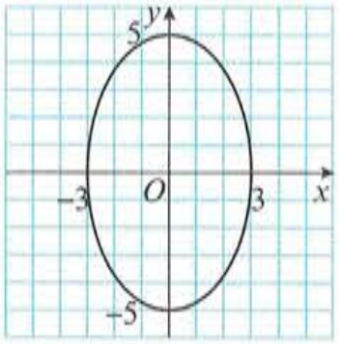
1.

t	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3
$x = 2t + 2$	-2	-1	0	1	2	3	4	5	6	7	8
$y = t^2$	4	2.25	1	0.25	0	0.25	1	2.25	4	6.25	9



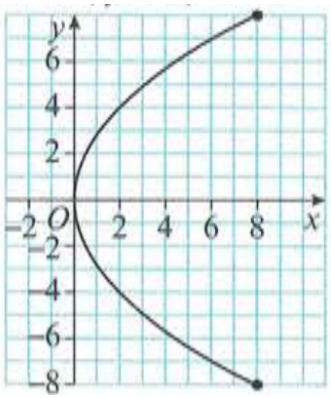
2.

t	0	$\pi/4$	$\pi/2$	$3\pi/4$	π	$5\pi/4$	$3\pi/2$	$7\pi/4$	2π
$x = 3\sin t$	0	2.12	3	2.12	0	-2.12	-3	-2.12	0
$y = 5\cos t$	5	3.54	0	-3.54	-5	-3.54	0	3.54	5

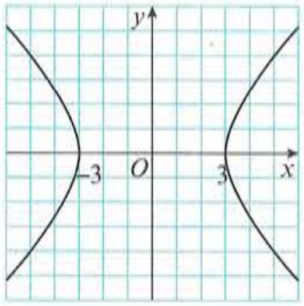


3.

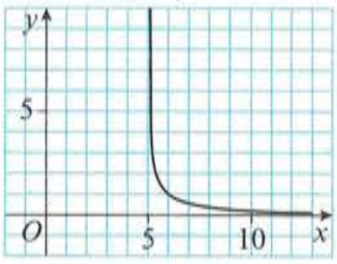
a.



b.



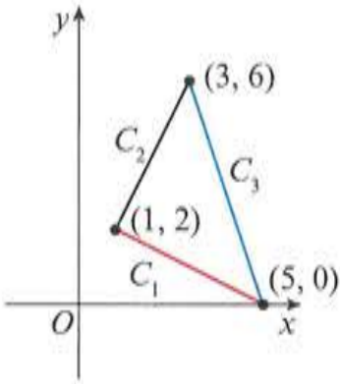
c.



4.

a. C_1: $y = 2.5 - 0.5x$, $1 \leq x \leq 5$
C_2: $y = 2x$, $1 \leq x \leq 3$,
C_3: $y = 15 - 3x$, $3 \leq x \leq 5$

b.

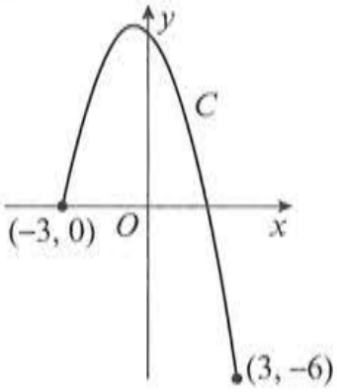


c. 10

5.

a. $y = (2 - x)(x + 3)$, $-3 \leq x \leq 3$

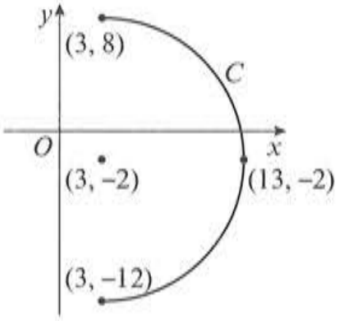
b.



6.

a. $(x - 3)^2 + (y + 2)^2 = 100$

b.



c. 10π

8.4 Points of intersection

1.

a. $(-7, 0)$, $(0, 7)$

b. $(-6, 0)$, $(3, 0)$, $(0, -6)$

c. $(-2, 0)$, $(30, 0)$, $(0, -3)$, $(0, 5)$

2.

a. $(1, 0)$, $(0, 2)$

b. $(-\pi, 0)$, $(\pi, 0)$, $(0, 1)$

c. $(\sqrt{2} - 3, 0)$, $(0, 7)$

3.

a. $t = -1$, $t = -6$

b. $(0, 2)$, $(-5, -3)$

4.

a. 3

b. $\dfrac{1}{2}$

5.

a. $t = 0$ and $t = \ln 3$

b. $(1, 0)$, $(9, 8)$

6.

a. $(0, 1)$ and $(-1, 1)$

b. Curve will intersect x-axis when $y = 0$

$t = \dfrac{7\pi}{12}$: $y = 2\sin\left(2\left(\dfrac{7\pi}{12}\right)\right) + 1 = 0$

$t = \dfrac{11\pi}{12}$: $y = 2\sin\left(2\left(\dfrac{11\pi}{12}\right)\right) + 1 = 0$

c. $\left(-\dfrac{\sqrt{6}+\sqrt{2}}{4}, 0\right)$, $\left(\dfrac{\sqrt{2}-\sqrt{6}}{4}, 0\right)$

7. $3y - 2\sqrt{3}x - 8\sqrt{3} = 0$

8.

a. $y = x - \ln 18$

$\Rightarrow \ln(t-1) = 2\ln(t+2) - \ln 18$

$\Rightarrow t - 1 = \dfrac{(t+2)^2}{18} \Rightarrow t^2 - 14t + 22 = 0$

$\Rightarrow t = 7 \pm 3\sqrt{3}$

b. $(\ln(9 + 3\sqrt{3}), \ln(6 + 3\sqrt{3}))$, $(\ln(9 - 3\sqrt{3}), \ln(6 - 3\sqrt{3}))$

8.5 Modelling with parametric equations

1.

a. 167 seconds (3 s.f.)

b. $y = 2.7 \times \dfrac{x}{3.6} = 0.75x$ which is a straight line.

c. $750\,\text{m}$

d. $4.5\,\text{m}\,\text{s}^{-1}$

e.

i. The path of the skier is unlikely to follow $y = 0.75x$.

ii. The speed of the ski lift is unlikely to be constant.

2.

a. 15 m

b. $16 - 7.6 \times \frac{17}{8} - \cos 17 = 0.125 = 12.5\,\text{cm}$ (3 s.f.)

c. 17 m

3.

a. 2.59 seconds (3 s.f.)

b. $k = 4.91$ (3 s.f.)

c. $x = 9\sqrt{2}t \Rightarrow t = \frac{x}{9\sqrt{2}}$

$y = -4.9\left(\frac{x}{9\sqrt{2}}\right)^2 + 9\sqrt{2}\left(\frac{x}{9\sqrt{2}}\right)$

$= -\frac{4.9}{162}x^2 + x$

4.

a. 0.429 seconds (3 s.f.)

b. 2.4 m

c. $0.8\,\text{m} \leq x \leq 1.6\,\text{m}$

5.

a. 7.45 cm (3 s.f.)

b. 17.7 cm (3 s.f.)

c. 8 cm

6.

a. $t = x - 5$, so $y = 5(x-5) - 2 = 5x - 27$
which is the equation of a straight line.

b. $\left(\frac{39}{5}, 12\right)$

c. Disagree; particles pass through point at different times.

7.

a. 391 m (3 s.f.)

b. 187 m, 578 m (3 s.f.)

c. 31 minutes

8.

a. 20 m

b. $k = \pi$, gradient $= 0.615$

Problem solving: Set A

Bronze

a.

i. $y = 5\sin 2t = 10\sin t\cos t = x\cos t$

ii. $\frac{x}{10} = \sin t$, $\frac{y}{x} = \cos t \Rightarrow \left(\frac{x}{10}\right)^2 + \left(\frac{y}{x}\right)^2 = 1$
$\Rightarrow 100(y^2 - x^2) + x^4 = 0$

b.

i. $(5\sqrt{2}, 5)$

ii. $k = 5 - 5\sqrt{2}$

Silver

a. $y = 4\cos\left(t + \frac{\pi}{3}\right) = 2\cos t - 2\sqrt{3}\sin t$

$= 2\cos t - 2\sqrt{3}\left(\frac{\sqrt{3}}{2}x\right) \Rightarrow y = 2\cos t - 3x$

$\Rightarrow \frac{3x+y}{2} = \cos t$

So $\left(\frac{\sqrt{3}}{2}x\right)^2 + \left(\frac{3x+y}{2}\right)^2 = 1 \Rightarrow (3x+y)^2 + 3x^2 - 4 = 0$

b.

i. $t = \frac{\pi}{6}$: $\left(\frac{\sqrt{3}}{3}, 0\right)$, $t = \frac{\pi}{2}$: $\left(\frac{2\sqrt{3}}{3}, -2\sqrt{3}\right)$

So gradient of line is $\frac{-2\sqrt{3}}{\frac{2\sqrt{3}}{3} - \frac{\sqrt{3}}{3}} = -6$

ii. $6x + y - 2\sqrt{3} = 0$

Gold

a. $y = 2\sin t + \sin 2t = 2\sin t + 2\sin t\cos t$

$= 2\sin t + 2\sin t\left(\frac{x-2}{2}\right) = x\sin t \Rightarrow \frac{y}{x} = \sin t$

$\Rightarrow \frac{y^2}{x^2} + \frac{(x-2)^2}{4} = 1 \Rightarrow y^2 = x^3\left(1 - \frac{1}{4}x\right)$, $2 - \sqrt{2} \leq x \leq 4$

b. $3 + 2\sqrt{2}$

c. $\{q : 1 \leq q \leq 3 + 2\sqrt{2}\}$

Problem solving: Set B

Bronze

a. 6.17, 19.1 (3 s.f.)

b. $t = \frac{x}{7.6}$

$y = -4.9\left(\frac{x}{7.6}\right)^2 + 16.3\left(\frac{x}{7.6}\right) = -\frac{245}{2888}x^2 + \frac{163}{76}x$

Height = 13.6 m (3 s.f.)

Silver

a. 32.1 m

b. $(0, -16)$

c. 1.57 minutes

Gold

a. $\ln\frac{9765625}{16}$

b. -0.675

Selected exam question bank answers

Question 10 Cartesian equation of curve is $y = 6x - 2x^2$

$3x + 2 = 6x - 2x^2 \Rightarrow 2x^2 - 3x + 2 = 0$

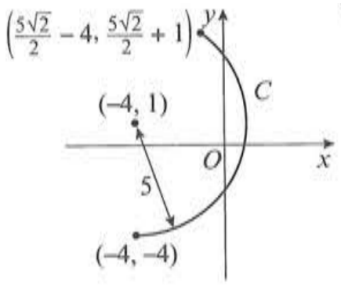
Using $b^2 - 4ac$: $(-3)^2 - (4 \times 2 \times 2) = -7 < 0$

So no solutions.

Question 31

a. $(x+4)^2+(y-1)^2=25$

b.



c. $\dfrac{15\pi}{4}$

Question 34

a. 1.9 m

b. $t = 0.66$ seconds

c. 2.51 m

Question 45

a. $y = 2x - 4x^4$, $0 \le x \le \sqrt{3}$

b. $t = \dfrac{1}{4}$. Range is $2\sqrt{3} - 36 \le \mathrm{f}(x) \le \dfrac{3}{4}$

Question 48

a. $A\left(-\dfrac{5}{3}, 0\right)$, $B\left(0, -\dfrac{5}{2}\right)$

b. $2y + 3x + 5 = 0$

Question 54

a. $x = \dfrac{1}{6}\cos t \Rightarrow 6x = \cos t \Rightarrow 216x^3 = \cos^3 t$
$y = \cos 3t = \cos(2t+t) = \cos 2t\cos t - \sin 2t \sin t$
$= (2\cos^2 t - 1)\cos t - (2\sin t \cos t)\sin t$
$= 2\cos^3 t - \cos t - 2(1-\cos^2 t)\cos t = 4\cos^3 t - 3\cos t$
$= 864x^3 - 18x = 18x(48x^2 - 1)$
So $a = 18$, $b = 48$, $c = 1$

b. $-\dfrac{\sqrt{3}}{12} \le x \le \dfrac{\sqrt{3}}{12}$

c. Turning points are $\left(-\dfrac{1}{12}, 1\right)$ and $\left(\dfrac{1}{12}, -1\right)$
Range is $-1 \le y \le 1$

Question 72

a. $(1, 4)$ and $(25, 28)$

b. $5y - 6x - 26 = 0$

Question 86

a. $T = 4$

b. $80\,\text{m}$, $t = 2$

c.

i. $82.9°$ (1 d.p.)

ii. The model is unlikely to be valid, as the answer to part i indicates the rollercoaster would virtually be heading directly into the ground.

d. $3410\,\text{m}^2$ (3 s.f.)